

4.8 Centers of Gravity Using Multiple Integrals

Motivation:

Now that we have the notion of double and triple integrals, we can easily get formulas for the center of mass of a lamina which does not have uniform density and center of mass of a three dimensional solid with varying density. We start by looking at the two-dimensional case.

Suppose that a lamina occupies a region D of the xy -plane and its density (in mass per unit area since we view the lamina as two dimensional) at a point in D is given by $\rho(x, y)$ where ρ is a continuous function on D .

Definition: The coordinates $(\bar{x}, \bar{y})$ of the center of mass of a lamina occupying the region D and having density function $\rho(x, y)$ are

$$\bar{x} = \frac{1}{m} \iint_D x \rho(x, y) dA \quad \text{and} \quad \bar{y} = \frac{1}{m} \iint_D y \rho(x, y) dA$$

Note: Three double integrals are required to find the center of mass of a lamina.

Note: Intuitively, one can view the double integrals for each coordinate of the center of mass as “weighted averages”.

Note: One will often hear $M_x = \iint_D y \rho(x, y) dA$ called the moment of the lamina about the x -axis, and will hear

$M_y = \iint_D x \rho(x, y) dA$ called the moment of the lamina about the y -axis.

Example 1: Suppose that a lamina occupies the region $D = \{(x, y) \mid y \leq 1 - x^2, y \geq 0\}$ and has density function $\rho(x, y) = y$. Find the center of mass of the lamina.

We now explore how to find the center of mass of a three dimensional solid with varying density.

Definition: Suppose that the density function of a solid object that occupies the region E is the continuous function $\rho(x, y, z)$ (the unit for this function is mass per unit volume).

The center of mass of this object is $(\bar{x}, \bar{y}, \bar{z})$ where $\bar{x} = \frac{1}{m} \iiint_E x \rho(x, y, z) dV$, $\bar{y} = \frac{1}{m} \iiint_E y \rho(x, y, z) dV$,

$\bar{z} = \frac{1}{m} \iiint_E z \rho(x, y, z) dV$, and $m = \iiint_E \rho(x, y, z) dV$.

Example 2: Find the center of mass of the solid that occupies the region bounded by the graphs of $x = y^2$, $x = z$, $z = 0$, and $x = 1$ which has density function $\rho(x, y, z) = 1$.